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On the Linear Superposition of Wing and Propeller Performance in a Wing Embedded Propeller System

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We continue our study on the aerodynamic performance of a Wing-Embedded Propeller (WEP) system by considering the effect of propeller installation location along the span and its overall size on the entire system. Six semi-span AR 2 wings with different spanwise opening locations and opening diameter-to-wing-chord ratios were investigated numerically using FlightStream® and experimentally in the University of Dayton Low-Speed Wind Tunnel without the installation of the propeller. The propeller is then installed and tested experimentally. Without the propeller, the wing with the opening closer to the wingtip experiences the least detriment to overall aerodynamic performance. Integrating the propeller into the WEP system and positioning it at the wing's mid-span, results in better overall performance. Increasing the propeller size has a more pronounced effect on the WEP system, increasing both lift and forward thrust generation. The linear superposition method is suitable for estimating the WEP system lift performance at a higher propeller incidence angle. However, it significantly underestimates the drag produced by the WEP system. This trend is consistent across all wing locations and propeller sizes tested in this study.

Nomenclature

AR	=	Aspect Ratio
b	=	Wing Span, (m)
c	=	Wing Chord, (m)
C_D	=	Drag Coefficient, $C_D = \frac{D}{qS_{Ref}}$
C_L	=	Lift Coefficient, $C_D = \frac{L}{qS_{Ref}}$
C_M	=	Moment Coefficient, $C_D = \frac{M}{qS_{Ref}}$
C_P	=	Pressure Coefficient, $C_P = \frac{P-P_\infty}{q}$
C_{Tp}	=	Forward Thrust Coefficient, $C_{TP} = \frac{T_P}{qS_{Ref}}$
D	=	Drag force, (N)
D_{Wing}	=	Wing Drag Force, (N)
d	=	Distance, (m)
d_r	=	Distance to the Wing Root, (m)
J	=	Advanced Ratio, $J = \frac{V_\infty}{nD}$
L	=	Lift force, (N)
L_{Prop}	=	Propeller Lift Force, (N)
L_{Wing}	=	Wing Lift Force, (N)
M	=	Pitching Moment, $(N.m)$
n	=	Propeller Rotational Speed, (RPS)
P	=	Power, (W)
q	=	Dynamic Pressure, (Pa)
Re	=	Reynolds Number
RPS	=	Rotations Per Second
S_{ref}	=	Reference area, (m^2)
T	=	Thrust Force, (N)
T_{Prop}	=	Propeller Forward Thrust, (N)
V_∞	=	Freestream Velocity, (m/s)
α	=	Angle of Attack, (Deg)
ρ	=	Air density, (kg/m^3)
θ	=	Propeller Incidence angle, (Deg)

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I. Introduction

Most modern Advanced Air Mobility (AAM) concepts integrate vertical takeoff and landing (VTOL) capabilities with fixed-pitch multi-rotors, distributed electric propulsion, and wing-embedded propulsion/propeller (WEP) systems [1]. Wing-embedded propulsion systems, considered since the 1960s in designs such as the Ryan XV5 by Ryan Aeronautical [2]were found easy to maintain and operate but were difficult to control, especially during the aircraft's transition and higher angle of attack operation.[3]. More recent design concepts include AgustaWestland's Project Zero which features a WEP system capable of rotating the propeller to provide thrust vectoring. However, it only managed to perform vertical takeoff and landing, and the transition to forward flight was never achieved [4].

Heyson [5] conducted wind tunnel experiments on a fan-in-wing concept aircraft. The model has a 1067mm wingspan with the NACA 16-017 airfoil and two 203.2mm diameter fans fixed in the midspan of the wing on each side. At small forward flight speeds, the loss in the fan-in-wing system lift generation is almost negligible. This indicates the possibility of linear superposition of independent wing and fan lift performances to estimate the system's overall lift generation.

Gregory et al. [6], Schaub [7], and Lieblein et al. [8] conducted a series of experiments to investigate and optimize the performance of the fan-in-wing system. They discovered that increasing the fan intake lip radius reduces flow separation and increases fan performance at higher forward flight speeds [6-7]. However, this would result in a higher back pressure loss in the fan wake [7]. An alternative is to reduce the fan inlet lip radius and install inlet vanes to prevent flow separation and minimize back pressure loss. Installing exit vanes on the lower surface of the wing also helps reduce back pressure loss in forward flight. However, this also reduced the overall lift generated by the fan-in-wing system [8].

Thouault et al. [9] conducted wind tunnel experiments and numerical investigation on a fan-in-wing model with a 1250mm wingspan, a NACA 16-020 airfoil, and two 70mm diameter fans placed 100mm from the wing root. An increment in lift coefficient, decrement in pitching moment coefficient, and a significant increment in drag coefficient were observed when operating the fan at higher forward flight speeds. The back pressure at the fan wake and the counter-rotating vortex pair were observed at the lower surface of the wing, due to the mixing of fan wake and freestream, resulting in a significant increase in drag.

Installing exit vanes can also help provide aircraft stability during the transition to forward flight. Hickey and Ellis [10] investigated a semi-span wing featuring an embedded fan with independent inlet and exit vanes using a NACA 16-015 wing profile. Our previous study [11] investigated the effect of a set of exit vanes on the performance of the WEP system with an Eppler 479 airfoil and APC 11x7 propeller. Wind tunnel tests indicate that exit vanes can generate net propulsive force at a range of freestream velocities with small reductions in lift at higher forward flight speeds. Various pitching moments were achievable by deflecting the exit vanes.

An alternative application of embedding the fan or propulsion system in the wing is to enhance aircraft stability. Deckert [2] and Kirk et al. [12] examined different WEP designs including large fans integrated into the wing for lift production and smaller fans integrated which provide roll and pitch moment. In most cases, placing a rotor in the wing resulted in a positive effect on lift generation. Placing the fan closer to the trailing edge of the wing can provide sufficient pitching and rolling moments with a minimal drag penalty [2,12]. However, placing the fan closer to the leading edge of the wing reduces the flow speed over the upper wing surface, resulting in significant lift penalties [2].

In our previous study [13], the changes in propeller performance when installed in a wing were investigated experimentally with a 0.28 m propeller diameter. An average of 20% loss in propeller streamwise thrust at propeller installation incidence angles (θ) between $55° < \theta < 90°$ was found due to the propeller wing interference. For the vertical lift force generation, a 13% loss was found at $\theta = 90°$. This loss diminishes when reducing θ to 70°. Further decrement in θ results in an increment in both forces, peaking at 17% at $\theta = 55°$. A subsequent study [14] investigated the overall aerodynamic performance of a wing-embedded propeller system using a 0.063m propeller installed in the center of a wing with a 0.254 m span and a 0.127 m chord, at angles between $60° < \theta < 90°$. The effect of the hole opening on the standalone wing could be modeled as a reduction in its effective aspect ratio. Using an empirical model based on the linear superposition of a standalone propeller and wing, the lift performance of the WEP system can be predicted within an angle of attack of $-12° < \alpha < 12°$ with an average error of less than 10% at $\theta \geq 75°$. However, our previous study [11, 13-14] only investigated WEP configurations when the propeller is placed at the center of the wing (mid-span and mid-chord) with a propeller diameter (D) to the wing chord (c) ratio of 0.5. To generalize the empirical model based on linear superposition developed in our previous study, the investigation of the WEP system with different propeller spanwise installation locations and diameter-to-wing-chord ratios is required. The current study aims to fill this knowledge gap and generalize the linear superposition model, which would benefit the preliminary design of a wing-embedded system for modern air-taxi concepts.

II. Experimental setup

All experiments were conducted in the University of Dayton's Low-Speed Wind Tunnel (UD-LSWT) in its open jet configuration. The tunnel has a 16:1 contraction ratio, 6 anti-turbulence screens, and a 76.2 cm by 76.2 cm open jet test section. The effective length of the test section is 182.9 cm. A 111.8 cm by 111.8 cm collector accepts the expanded air on its return to the diffuser. A pitot tube is connected to a TSI T600 Micromanometer which was used to measure the freestream velocity during the experiment. The experimental setup used for the standalone wing studies, as well as the combined system of the propeller and wing system is shown in Figure 1.

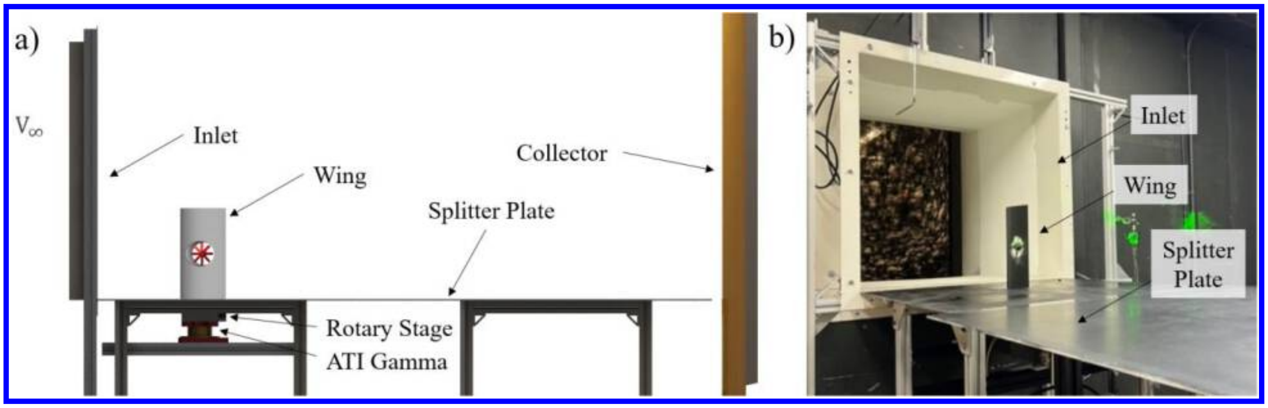

Figure 1. a) Side View CAD Schematic and b) Isometric View Picture of UD-LSWT experimental setup

A. Standalone Wing Test Setup

Nine different standalone wing configurations without an installed propeller are shown in Figure 2, and the corresponding test matrix is shown in Table 1. Each wing was designed using the Eppler 479 airfoil, with a half-span aspect ratio of 2.0 and a circular hole opening diameter of 0.064 m located at the mid-chord of the wing for propeller installation. A lip with a 0.5 width-to-length ratio was added to the circular opening to improve the propeller's inflow, which is consistent with the model in our previous study [13]. Two different propeller diameter-to- -chord ratios (D/c) of 75% (0.0846 m chord) and 50% (0.127 m chord) were selected for the experiment which represents a realistic propeller sizing ratio for modern AAM concepts. Three different propeller spanwise installations (d_r/b) locations were tested: 0.25, 0.5, and 0.75 d_r/b.0.25 d_r/b represents the opening location closer to the wing root and 0.75 d_r/b represents the opening location closer to the wingtip. The wings were mounted to a rotary stage and an ATI Gamma F/T balance [14] assembly.

A 0.91m x 1.83m splitter plate was used to achieve a half-span configuration of the models tested. The tunnel speed for the no-propeller wing cases was set at 15 and 25 m/s. The angle of attack varied from -20 to 20 degrees with 1 degree increment. The F/T data was sampled at the rate of 1000 Hz for a 15-second duration at each angle of attack. Two trials were run in each case to ensure data repeatability. To minimize sensor drift, tare values with the propeller powered off were recorded before and after each test. The average of these two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare value was greater than 0.1N of force on any axis, the experimental data was then rejected, and the experiments were conducted again.

B. Wing-Embedded Propulsion Test Setup

The test matrix for the wing-embedded propeller (WEP) system is shown in

Table 2. A T-motor T-2004 brushless motor and an eight-blade 2.5'x1.5' (63mm diameter and 38 mm pitch) propeller were selected to be consistent with our previous study [13]. The propeller rotational speeds were set at 625 RPS for all cases, with freestream velocities ranging from 0 to 20 m/s, resulting in an advance ratio between 0 and 0.52. All three wings with a D/c of 50%, shown in Figure 2, are selected for the WEP system to investigate the effect of propeller spanwise location on the overall WEP system performance. For D/c of 75%, only one wing (0.75 d_r/b) was selected based on its standalone performance to assess the effect of propeller size on the WEP system performance. The WEP system was tested at a velocity range of 10 m/s to 20 m/s, which is equivalent to the propeller advance ratio of 0.26 and 0.52. Tests were conducted at angles of attack between -15 and 15 degrees with a 1-degree increment. The F/T data was sampled at the rate of 1000 Hz for a 15-second duration at each angle of attack. A Finite Impulse

Response (FIR) band-stop filter [16] was applied to the data to filter out the vibration frequency caused by the propeller rotation and the natural frequency of the aluminum test frame. The tap tests on the test stand were performed with the propeller powered off which resulted in the structure's natural frequency of 45Hz.

Table 1. Test Matrix of the Standalone Wing Experiment

D/c	Reference Area(m^2)	Wing Reynolds Number (x1,000)	Opening Spanwise Location (d_r/b)	Wing angle of attack (degrees)	Freestream Velocity (m/s)
75%	0.01140	98 ~ 163	0.25, 0.5, 0.75	-20 to 20	15, 25
50%	0.02932	147 ~ 245			

Table 2. Test Matrix of the Wing Embedded Propeller Experiment

D/c	Opening Spanwise Location (d_r/b)	Wing angle of attack (degrees)	Freestream Velocity (m/s)	Propeller Rotational Speed (RPS)	Propeller Mounting Incidence Angle (degrees)
75%	0.75	-15 to 15	10,15,20	625	60, 75, 90
50%	0.25, 0.5, 0.75				

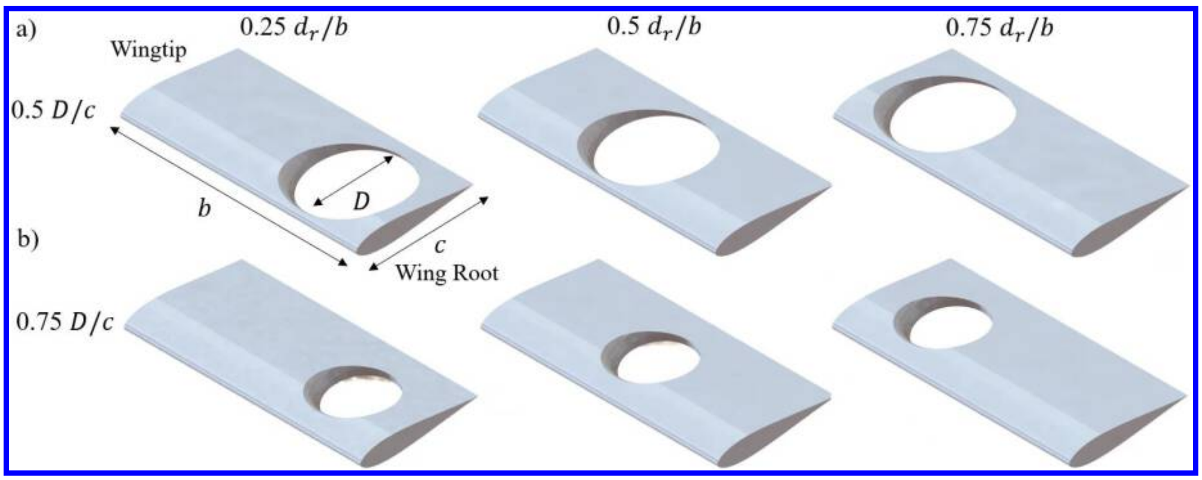

Figure 2. Schematic of the Different Wing Configurations Studied: a) 75% D/c and b) 50% D/c

Figure 3 shows the schematic of the WEP system assembly. The propeller and motor are mounted on a motor interface, which is secured to the wing body. The propeller mounting incidence angle (θ) is defined as the angle between the freestream and the propeller axial direction when the WEP system is at $\alpha = 0°$ as annotated in the figure. Hence, $\theta = 90°$ closely resembles the propeller edgewise flight condition which represents the WEP system under the VTOL phase for a rotating propeller design, or the forward flight condition for a vehicle design with a pusher propeller configuration. The $\theta = 60°$ represents the propeller tilting forward under forward flight conditions. To change the propeller incidence angle, the mounting interface is replaced. The interface is designed to keep the center of the propeller hub at the upper surface of the wing, which is consistent with our previous study [12,13].

D. Numerical Setup with FlightStream®

FlightStream® [17] was used for numerical investigation of the standalone wing in various configurations. FlightStream® features two main types of potential flow solvers: a pressure-based solver and a vorticity solver. The pressure-based solver is used in the present study, which utilizes the pressure fields on the model geometry to determine the aerodynamic loads. A combination of source and sink panels along with doublet panels on the trailing edge is used to provide a solution for pressure-based potential flow solvers.

The surface mesh of the wing and the splitter plate in FlightStream® is shown in Figure 4. The wing platform has a structured mesh with 80 spanwise node points and 60 chordwise node points on both the upper and lower surfaces.

The chordwise mesh also has a dual side growth rate of 1.05 to better resolve the leading and trailing edge geometry. An unstructured mesh is used for the circular hole to better represent the complex geometry. Moreover, the splitter plate of the same size as the one used in the wind tunnel experiment is included in the model to better simulate the actual experimental conditions. The splitter plate has 40 node points from side to side and 60 node points, with a dual-side growth rate of 1.05 downstream.

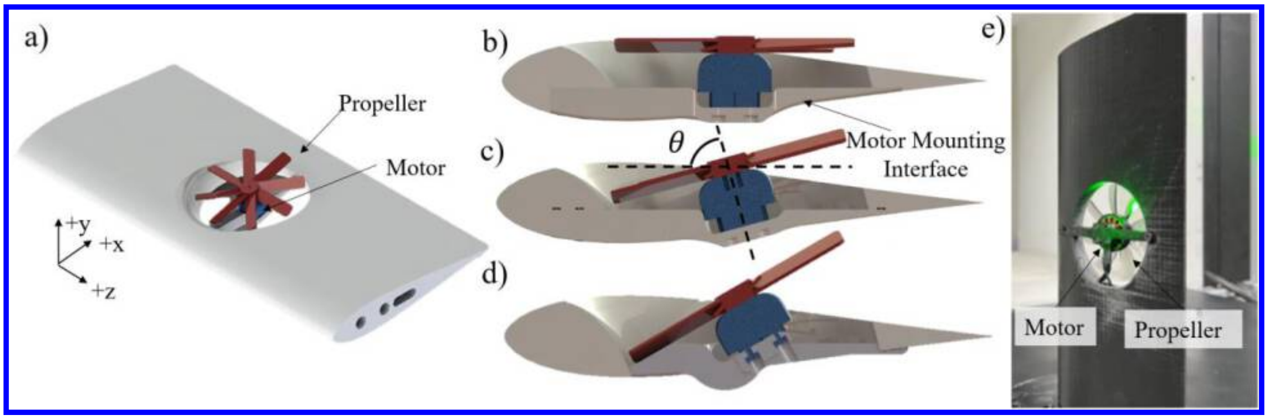

Figure 3. Schematic Showing the Wing-Embedded Propeller Configuration with a) Isometric View b) $\theta = 90°$, c) $\theta = 75°$, d) $\theta = 60°$ and e) Picture of the Tested Model

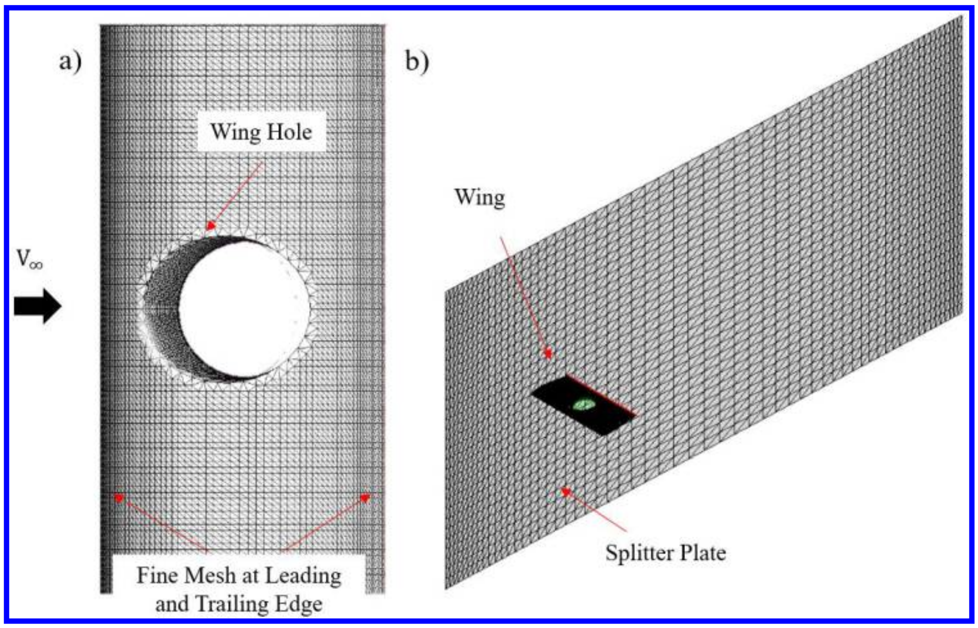

Figure 4. Surface mesh in FlightStream® for a) circular hole wing and actuator propeller and b) wing and splitter plate

III. Experimental Results

The first sub-section discusses the experimental results from the standalone wing performance, the test matrix of which is shown in Table 1. And the second subsection focuses on the experimental performance results from the different WEP systems, the test matrix for which is shown in Table 2. The final sub-section discusses the application of the linear superposition model.

A. Standalone Wing Performance

The experimental and numerical coefficient of lift (C_L) vs. wing angle of attack (α) for the standalone wing cases for $D/c = 0.5$ and $D/c = 0.75$ cases is shown in Figure 5. The experimental results are represented as scatter points, while the FlightStream® results are shown as dashed lines. The surface area of each wing shown in Table 1 was used to normalize the lift and drag forces to obtain lift and drag coefficients. For the experimental results, the C_L of all wings with the holes has a lower lift curve slope than the baseline due to the circular opening on the wing. Different lift curve slopes (dC_L/d_α) are also observed among different d_r/b and D/c cases. Based on the McCormick formula [18], the calculated effective AR based on the dC_L/d_α for the baseline wing at $D/c = 0.5$ and $D/c = 0.75$ is shown in Table 3.

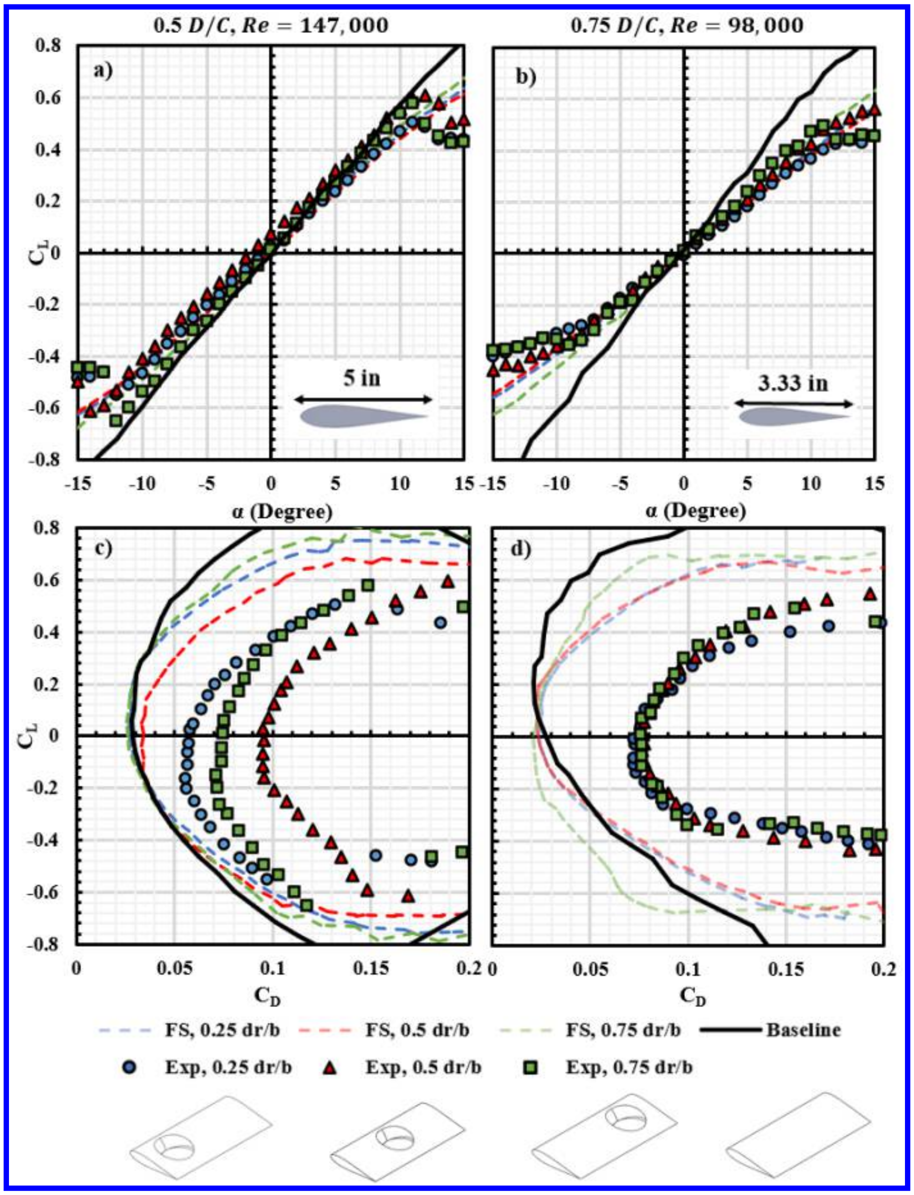

Figure 5. a) - b) C_L vs. α and c) - d) C_L vs. C_D for $D/c = 0.5$ and $D/c = 0.75$ Standalone Wing

$$a = \frac{dC_L}{d\alpha} = a_0\left(\frac{AR}{AR + 2\left(\frac{AR+4}{AR+2}\right)}\right) \tag{1}$$

According to the experimental data of both D/c cases, the effect of the hole locations is similar across different chord lengths. The $d_r/b = 0.75$ shows the highest dC_L/d_α while the lowest dC_L/d_α is observed at $d_r/b = 0.25$. The calculated dC_L/d_α is shown in Table 3 below. 0.5 D/c has a higher dC_L/d_α than 0.75 D/c for all cases, and the trend across hole location is consistent. This trend is also predicted through FlightStream® within $-11 \leq \alpha \leq 11$. Above this α range, stall is observed in the wind tunnel experimental results, while FlightStream® does not predict stall until $\alpha \sim 17°$. For the effect of the wing opening size, $D/c = 0.5$ shows a considerably higher dC_L/d_α than $D/c = 0.75$, resulting in a calculated effective aspect ratio of 2.6 at $d_r/b = 0.25$ for its best performer. Moreover, $D/c = 0.5$

wings show a sharp stall at $\alpha \sim 12°$ while a more gradual stall performance is observed for the $D/c = 0.75$ cases at the similar α range. FlightStream® is able to predict the change in $dC_L/d\alpha$ between chord lengths and hole locations for C_L performance with an error less than 5% as seen in Table 3. However, it does not account for stall for either case shown in Figure 5a and b; this is expected as it underestimates flow separation around the wing opening region.

Table 3. Calculated Effective AR for Standalone Wing Cases from Experiment and FlightStream®

	$0.25\ dr/b$		$0.5\ dr/b$		$0.75\ dr/b$	
	$AR_{Eff\ Exp}$	$AR_{Eff\ FS}$	$AR_{Eff\ Exp}$	$AR_{Eff\ FS}$	$AR_{Eff\ Exp}$	$AR_{Eff\ FS}$
$0.5\ D/c$	2.03	2.14	2.16	2.12	2.60	2.66
$0.75\ D/c$	1.56	1.61	1.72	1.68	2.03	2.12

Next, the drag polar (represented as C_L vs. C_D) of the standalone wing configurations are shown in Figure 5c and d. A larger difference in hole location is observed between all three wings with $D/c = 0.5$ than $D/c = 0.75$ cases. For $D/c = 0.5$, the $d_r/b = 0.50$ wing produces the most drag while $d_r/b = 0.25$ produces the least amount of drag. The $D/c = 0.75$ case does not display these discrepancies, as they do not experience a large difference in drag. As mentioned previously, the difference between the FlightStream® data and the experimental data is explained by FlightStream® not accounting for flow separation around the wing opening region. In this case, the results from FlightStream® underestimate the drag generated by the wing, while still preserving the overall trend between different d_r/b cases.

Although FlightStream® underestimated C_D for all cases, it was able to accurately predict C_L for all standalone wing cases as shown in Figure 5. The wing upper surface pressure coefficient (C_P) contour for all cases tested at $\alpha = 8°$ is shown in Figure 6. The freestream comes from the left-hand side and the wing root is at the bottom of the wing as annotated. For both baseline cases without an opening, a lower C_P distribution is observed closer to the wing root. The overall magnitude of C_P increases closer to the tip, due to the effect of the wingtip vortex. For different D/c, a lower C_P is observed closer to the leading edge for the $D/c = 0.75$ wing baseline, along with a slightly higher C_P closer to the trailing edge, as a result of the Reynolds number effect. It is clear that adding a circular opening affects the local pressure distribution on the wing. In almost all cases, the suction is preserved upstream of the opening. A sudden increment in C_P is then observed as the flow enters the opening due to flow separation at the lip. For different d_r/b, the circular opening closer to the wingtip ($d_r/b = 0.75$) has the least disruption to lift generation when compared to other cases since lift distribution tapers to zero closer to the tip. This results in a higher overall lift generated when compared to other cases. A similar trend is seen for the 0.75 D/c case.

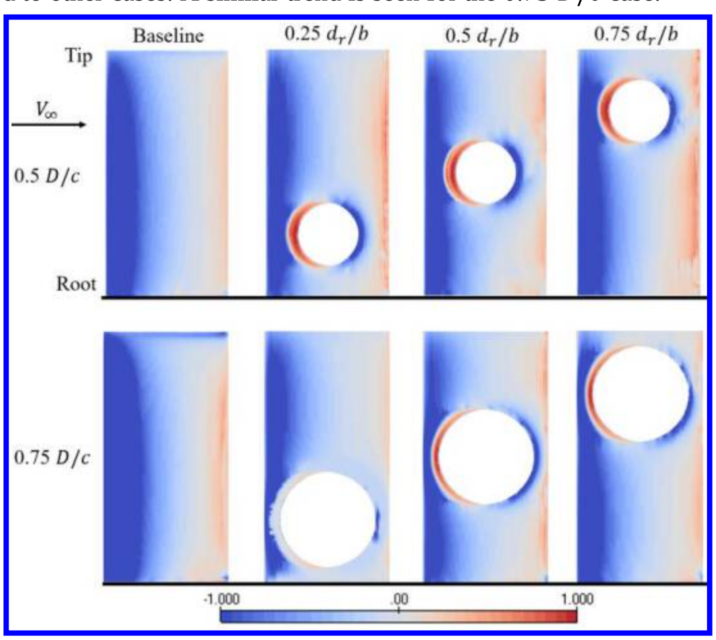

Figure 6. FlightStream Estimated Wing Upper Surface C_P Contour for All Cases Tested at $\alpha = 8°$.

B. Various Wing Embedded Propeller System Performance

For the wing-embedded propulsion system testing, the propeller rotational speed was held at 625 RPS at three different V_∞ of 10, 15, and 20 m/s corresponding to three different advance ratios J of 0.25, 0.38, and 0.51 for the propeller, respectively. The results are shown with respect to the wing reference coordinates in terms of lift and drag.

B.1 Effect of Wing Opening Spanwise Locations on WEP System Performance

The C_L vs. α results from the WEP at different freestream velocities for $D/c = 0.5$ are shown in Figure 7a,b and c. Each color represents the hole location (d_r/b) while markers corresponding to the propeller pitch angle (θ) with respect to the wing.

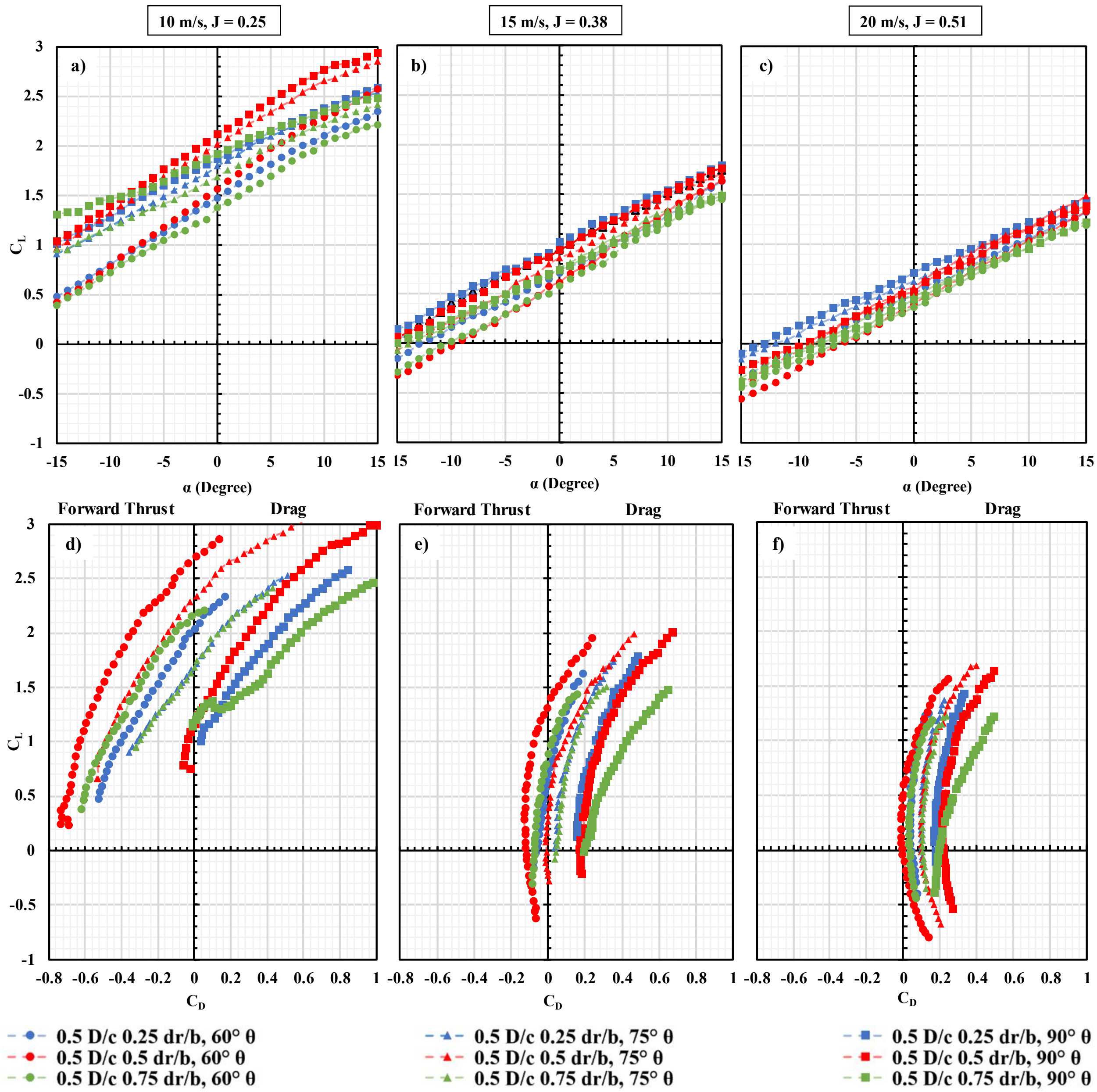

Figure 7. a) – c) C_L $vs. \alpha$ and d) - f) C_L vs. C_D for WEP $D/c = 0.5$ at 10 m/s, 15 m/s, and 20 m/s Freestream Velocities

It is evident that the effect of the propeller on lift generation of the WEP system is more significant at lower speeds than at higher speeds due to the lower advance ratio. Consistent at every freestream velocity, the propeller at $\theta = 60°$

creates the least amount of lift, whereas the propeller at $\theta = 90°$ typically creates the most lift. On the other hand, as θ increases, dC_L/d_α also increases, as shown in Figure 4. Additionally, the effect of d_r/b on the WEP C_L shows a dependence on advance ratio. $d_r/b = 0.25$ generates comparatively more lift at higher J while the $d_r/b = 0.50$ generates comparatively more lift at lower J. Placing the propeller closer to the wingtip ($d_r/b = 0.75$) tends to have the lowest benefit on lift generation when compared to the ones that are mounted inboard.

Table 4. Calculated dC_L/d_α (deg^{-1}), for all $D/c = 0.5$ cases at $V_\infty = 10$ m/s

	0.25 dr/b	0.5 dr/b	0.75 dr/b
$\theta = 60°$	0.0683	0.0773	0.0655
$\theta = 75°$	0.0600	0.0654	0.0555
$\theta = 90°$	0.0541	0.0704	0.0513

The drag polar is shown in Figure 7d e and f. It is worth noting that when C_D values are negative, the WEP system is essentially generating forward thrust. Similar to C_L, the propeller has a higher effect on the WEP system thrust production at a lower V_∞. For all V_∞ tested, the propeller at 90 degrees consistently creates the most drag while the propeller at 60 degrees provides the least amount of drag and the most amount of forward thrust. Additionally, the effect of d_r/b on WEP C_D changes depending on the propeller pitch angle (θ). At $\theta = 90°$, the $d_r/b = 0.75$ case creates the most drag whereas the $d_r/b = 0.25$ creates the least drag. However, at $\theta = 60°$, the $d_r/b = 0.5$ case produces the least drag (most forward thrust) while $d_r/b = 0.25$ produces the most drag (least forward thrust). A key observation is that positioning the propeller at θ=90° does not significantly enhance lift generation in the WEP system compared to θ=75°. However, the advantage in forward thrust generation at θ=75° is much more substantial. With similar lift being generated at both 90° and 75°, but with significantly higher drag at 90° compared to 75°, it is advantageous to install the propeller at a 75° angle to the wing rather than at 90°, especially at lower advance ratios.

B.2 Effect of Propeller-to-Wing Ratio on WEP System Performance

In this section, the effect of the propeller-to-wing ratio (D/c) on the WEP system performance will be discussed. The WEP systems with 0.5 and 0.75 D/c and $d_r/b = 0.75$ is selected and the C_L vs. α results at 10.25 J and 0.51 J are shown in Figure 8a and b. At 0.25 J , $D/c = 0.75$ cases boast a much higher C_L than the $D/c = 0.5$ cases at all θ tested. However, when increasing J, the difference in C_L reduces. It is evident that the large increase in C_L for $D/c = 0.75$ WEP system is a direct result of the propeller size increment with respect to the chord. A larger D/c leads to a much larger propeller influence on the WEP system resulting in a much higher C_L increment. Similar to the previous section, both $\theta = 90°$ case produces a higher C_L under both J. And with the decrement of θ, the C_L produced decreases. Meanwhile, the lift curve slope for all cases increases when reducing θ. As a result of this, the effective aspect ratio of the WEP system increases, especially for the $V_\infty = 10$ m/s cases where the lift curve slope almost doubled at $\theta = 60°$ when compared to $\theta = 90°$, which is shown in Table 5.

Figure 8c and d show the drag polar. It is clear the $D/c = 0.75$ cases outperform their $D/c = 0.5$ counterpart at both J as a result of a higher propeller influence on the WEP system for both C_L and C_D.. On the other hand, a lower drag (or higher forward thrust) is observed at $\theta = 60°$ case when compared to the higher θ cases as a result of the propeller forward pitch. When increasing J, the forward thrust generated by the WEP system reduces. At 0.51 J, only $D/c = 0.75$, $\theta = 60°$ cases were capable of generating forward thrust, while all other WEP systems would require a further reduction in θ to generate the thrust for forward flight.

Table 5. Calculated dC_L/d_α (deg^{-1}), for $D/c = 0.5$ and 0.75 at $V_\infty = 10$ m/s and 20 m/s

	$\theta = 60°$		$\theta = 75°$		$\theta = 90°$	
	10 m/s	20 m/s	10 m/s	20 m/s	10 m/s	20 m/s
0.5 D/c	0.065	0.061	0.058	0.057	0.045	0.047
0.75 D/c	0.086	0.059	0.054	0.055	0.040	0.042

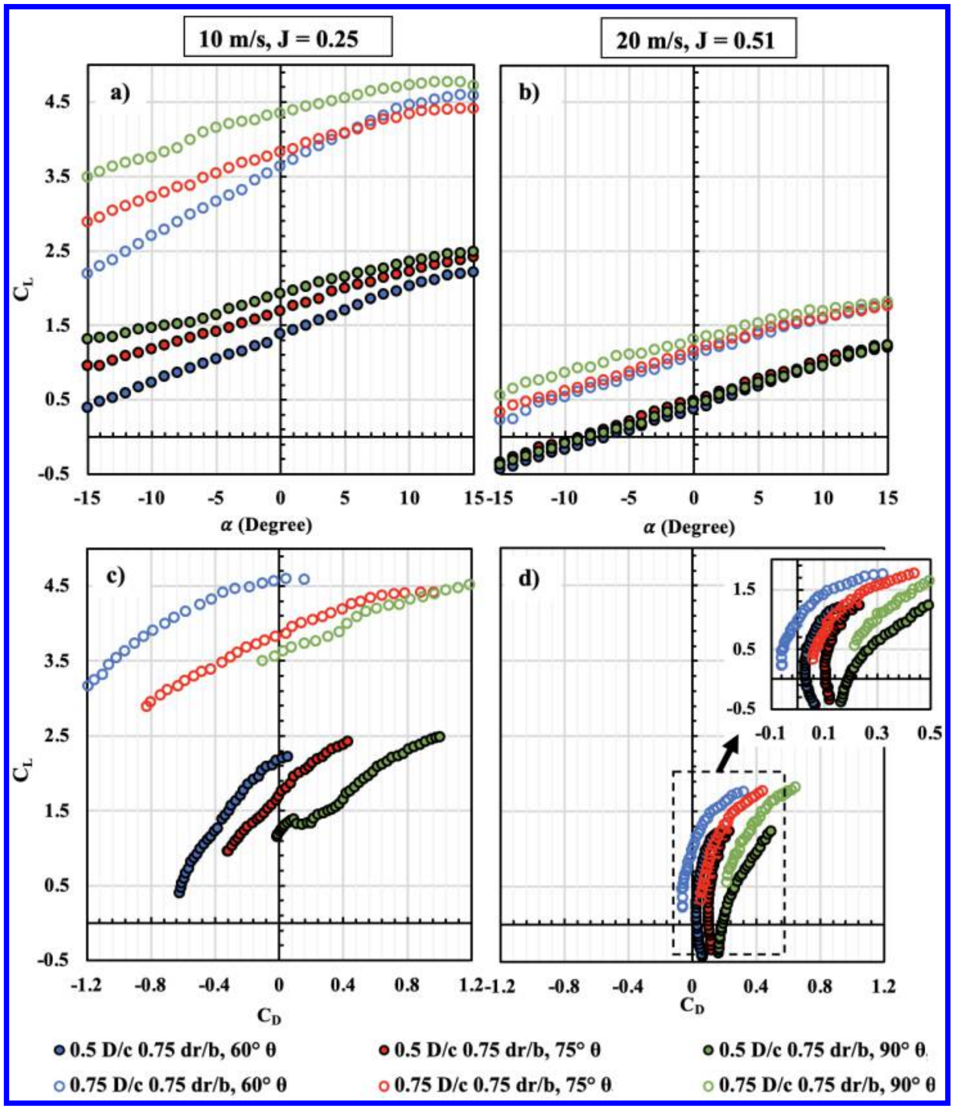

Figure 8. a) – b) C_L vs. α and c) – d) C_L vs. C_D for 0.5 amd 0.75 D/c wings at $0.75\ dr/b$ at different advance ratios

C. Application of the Linear Superposition Model on Wing Embedded Propeller System

To evaluate if the linear superposition model can be applied to estimate the performance of the WEP system, the summation of the individual performance of the standalone wing (Figure 5) and the performance of the standalone propeller (from our previous study [13]) should be equal to the measured WEP system performance. The summation of both standalone performances is also the ideal condition where no wing-propeller interference occurs, which can be expressed as:

$$L_{ideal} = L_{Prop} + L_{Wing} \qquad D_{ideal} = -T_{Prop} + D_{Wing} \tag{2}$$

where L_{Prop} is the propeller thrust on the wing normal (lift) direction and T_{Prop} is the propeller thrust in the wing axial (freestream) direction, which is shown below in Figure 9.

Therefore, the ideal lift and drag coefficient is calculated accordingly:

$$C_{L_{Ideal}} = \frac{L_{Prop} + L_{Wing}}{qS_{Ref}} \qquad C_{D_{Ideal}} = \frac{-T_{Prop} + D_{Wing}}{qS_{Ref}} \tag{3}$$

As discussed in our previous study [13], the influence of the propeller on the WEP system will be higher when operating at a lower forward flight speed or at a higher D/c, where the lift generated by the wing is significantly lower than the thrust generated by the propeller. Hence, to eliminate such effects, the total thrust generated by the propeller $C_{T_{total}}$ at the same V_∞ is used to normalize the C_L and C_D difference between the ideal condition and the WEP system as:

$$\Delta C_{L_{Norm}} = \frac{C_{L_{WEP}} - C_{L_{Ideal}}}{C_{T_{Total}}} * 100 \qquad \Delta C_{D_{Norm}} = \frac{C_{D_{WEP}} - C_{D_{Ideal}}}{C_{T_{Total}}} * 100 \tag{4}$$

If we assume that the performance of the wing remains constant, then $\Delta C_{L_{Norm}}$ and $\Delta C_{D_{Norm}}$ represent the changes in propeller performance when installed on the WEP wing. These changes account for all wing-propeller interference in terms of $\Delta C_{L_{Norm}}$ and $\Delta C_{D_{Norm}}$. A positive $\Delta C_{L_{Norm}}$ indicates a higher lift measured for the WEP system compared to the ideal condition, while a positive $\Delta C_{D_{Norm}}$ signifies a higher forward thrust.

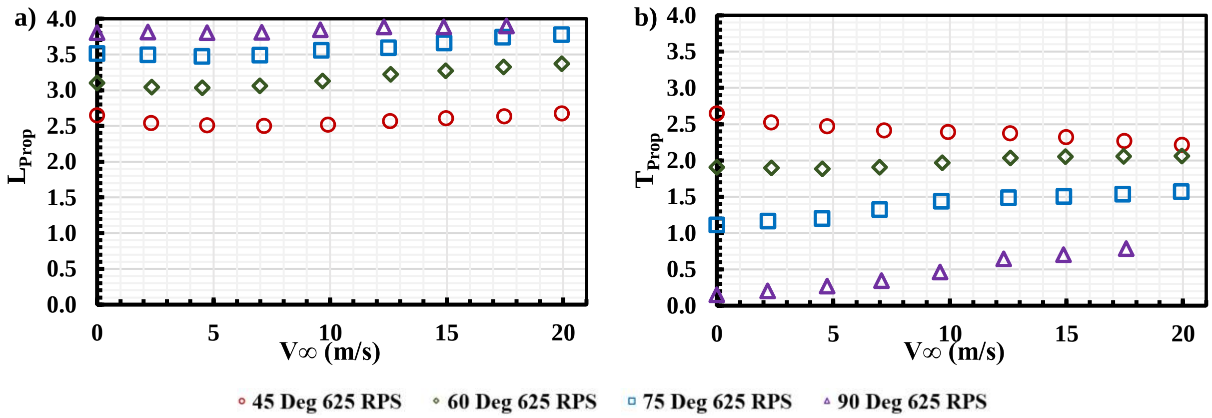

Figure 9. L_{Prop} and b) T_{Prop} under different θ vs. V_∞

C.1 Linear Superposition on Lift Performance

Figure 10 shows $\Delta C_{L_{Norm}}$ $vs. \alpha$ for all 0.5 D/c WEP systems at J of 0.25 and 0.51. The discussion will focus on $-10° < \alpha < 10°$ for practical applications. 0.25 J, the WEP system exhibits relatively small losses on lift at $\theta = 90°$ and 75° indicating an almost negligible interference between the propeller and the wing on lift generation. Figure 10a indicates that at 0.25 J, the performance of the WEP system is predictable with 10% error through the linear superposition method for cases where the propeller location is at 0.5 and 0.75 d_r/b. The case with propeller location at 0.25 d_r/b showed an error of around 20% from the ideal scenario indicating a negative propeller interference. The 20% error holds true for most cases at higher J as shown in Figure 12b except for the 0.25 d_r/b case when the propeller θ is at 75° angle. It is however surprising to see a positive propeller interference for all θ =60° inclination cases at higher J . It is hypothesized that the orientation of the propeller disk at $\theta = 60°$ generates an additional pressure difference between the upper and lower surface of the wing leading to an increased performance when compared to the ideal case. Hence, for this particular case, it is safe to apply the linear superposition method to the lift estimation, though it is necessary to keep in mind that the resulting lift will be slightly higher than the ideal condition. For the other cases, the linear superposition holds with a 10-20% error margin.

Figure 11 shows the results for $D/c = 0.75$ and $D/c = 0.5$ wing at various θ and J. At 0.25 J, $D/c = 0.75$ shows a smaller $\Delta C_{L\,Norm}$ when compared to $D/c = 0.5$ at all θ. At higher J a positive $\Delta C_{L\,Norm}$ is observed for all θ cases at 0.75 D/c indicating a positive propeller-wing interaction and a higher propeller efficiency at larger D/c. The overall changes in $\Delta C_{L\,Norm}$ are less than 8% at both V_∞ for the 75% D/c, indicating the linear superposition method is still applicable, if not better, for a larger propeller on the WEP system.

C.2 Linear Superposition on Thrust/Drag Performance

The results for $\Delta C_{D\,Norm}$ are displayed in Figure 12 for all variances of $D/c = 0.5$ at both advance ratios. It is clear that all cases demonstrate a higher loss in the WEP system forward thrust generation, or in other words, a higher drag experience by the WEP system from the ideal scenario. Therefore, the linear superposition method might not be suitable for estimating the thrust or drag generated by the WEP system. At 0.25 J, $\Delta C_{D\,Norm}$ remains relatively constant for all $\theta < 90°$ cases, while at 0.51 J, the magnitude $\Delta C_{D\,Norm}$ reduces with the increment of α for almost all cases. For the $\theta = 90°$ case, the slope of $\Delta C_{D\,Norm}$ vs. α is higher than other cases at both V_∞. Among all three wings, the $D/c = 0.5$ WEP system shows the lowest increment in drag production, indicating a better overall performance.

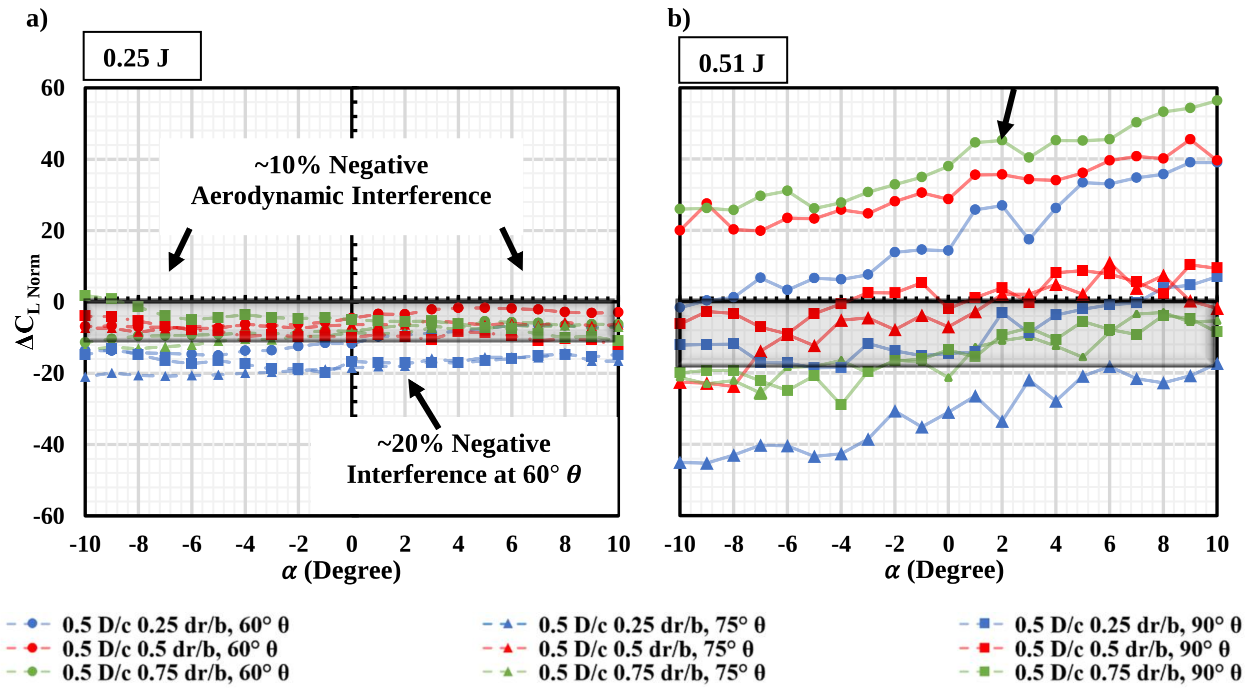

Figure 10. $\Delta C_{L\ Norm}$ vs. α for all 0.5 D/c cases at a) 0.25 J and b) 0.51 J

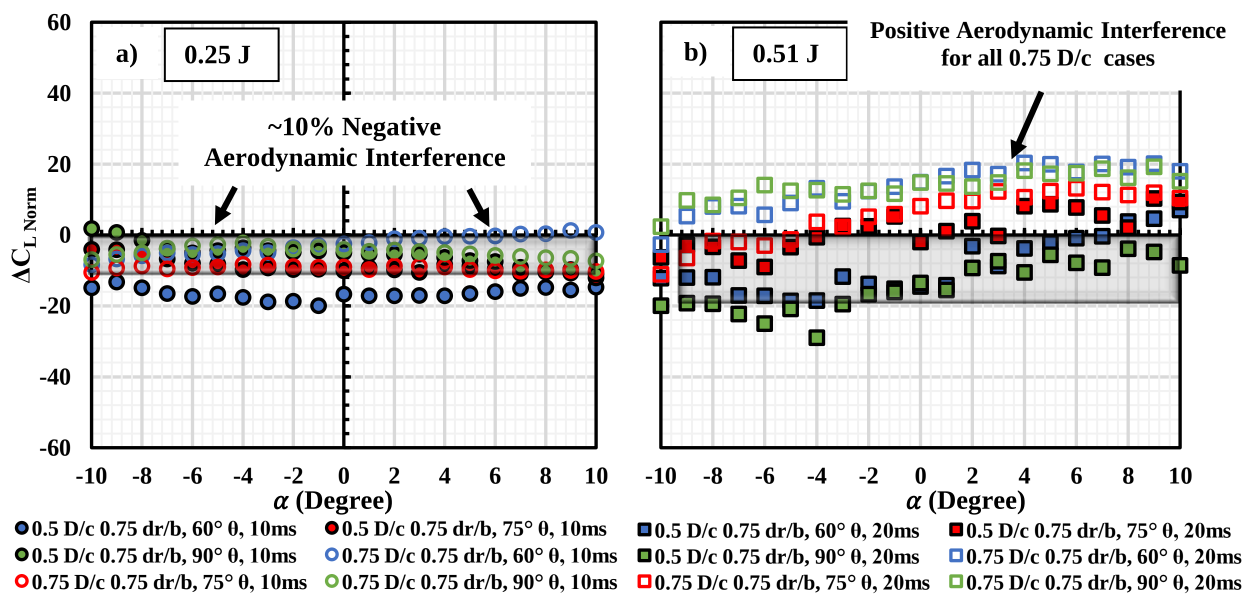

Figure 11. $\Delta C_{L\ Norm}$ vs. α for both 0.75 d_r/b cases at a) 0.25 J and b) 0.51 J

The results for $\Delta C_{D\ Norm}$ with different D/c ratios are shown in Figure 13. A similar behavior in magnitude and trend as seen in Figure 12 is observed in Figure 13 $\Delta C_{D\ Norm}$ as well for both Js. This indicates a negative effect of the propeller and wing interaction. Again, the linear superposition method is not suitable for estimating the drag for the WEP system. When placing the propeller in the WEP system, the larger drag production than the ideal condition is likely due to massive flow separation underneath the wing as a result of the propeller wake, which leads to a higher momentum deficit. A detailed flow analysis method such as particle image velocimetry (PIV) is required to further examine the flowfield and understand the effect of the propeller and wing interaction.

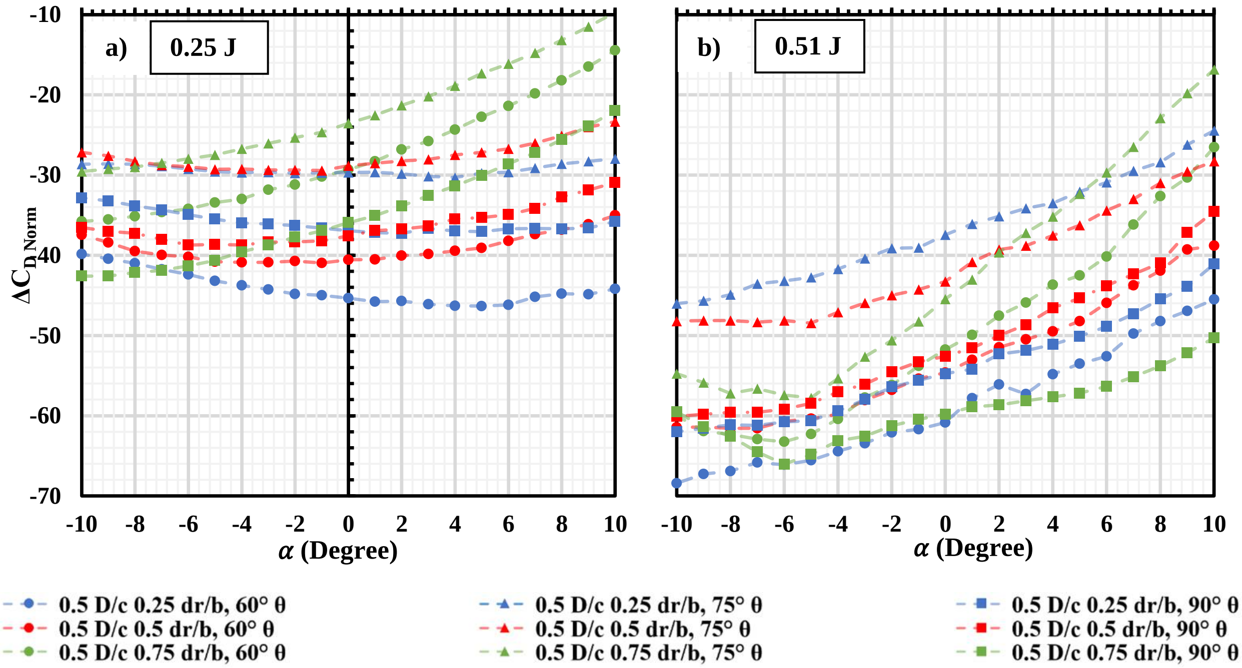

Figure 12. $\Delta C_{D\ Norm}$ vs. α for all 0.5 D/c cases at a) 10 m/s and b) 20 m/s

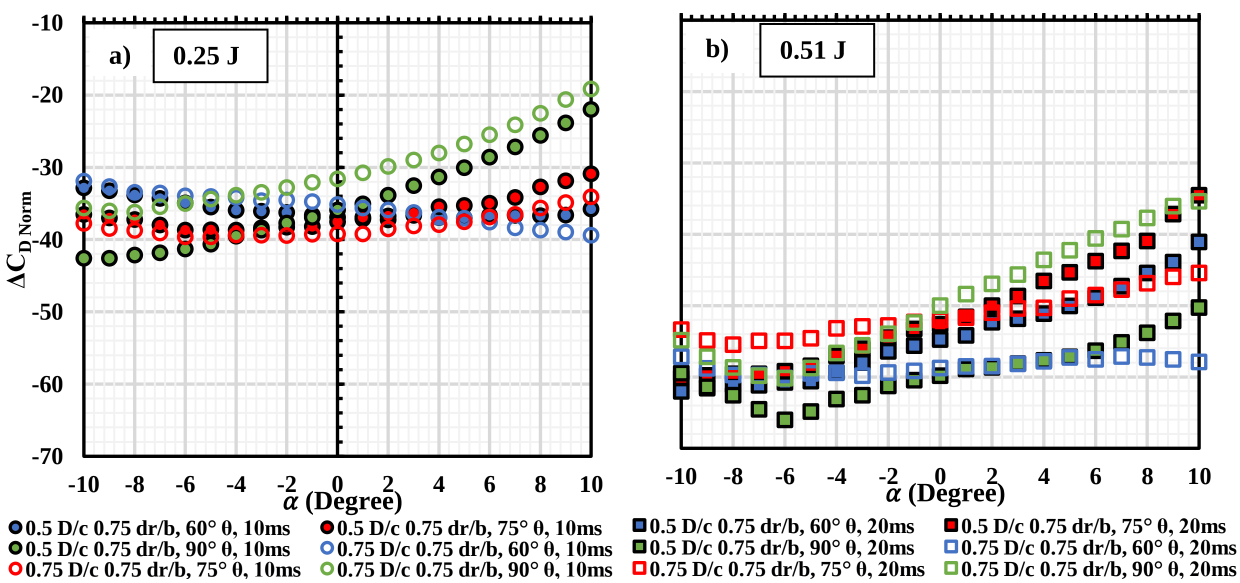

Figure 13. $\Delta C_{D\ Norm}$ vs. α for both 0.75 d_r/b cases at a) 10 m/s and b) 20 m/s

IV.Conclusion

The Wing Embedded Propeller (WEP) system was experimentally investigated, focusing on different opening locations with respect to the wingspan and overall sizes of the opening with respect to the chord. For the standalone wings without the installation of the propeller, adding a circular opening on the wing results in a reduced lift curve slope, which can be modeled by a lower effective aspect ratio. Placing the opening closer to the wingtip has the least penalty on the lift generation. This trend preserves when increasing the size of the opening. However, a larger opening also leads to a higher loss in the lift, resulting in an even lower effective aspect ratio.

When a propeller is installed on the WEP system, the impact of the circular opening and propeller installation location varies with forward speed and advance ratio. At lower advance ratios, positioning the propeller at the mid-span improves the lift generated by the WEP system. As the advance ratio increases, placing the propeller closer to the wing root shows benefits.

Positioning the propeller at θ=90° does not significantly enhance lift generation in the WEP system compared to θ=75°. The advantage in forward thrust generation at θ=75° is much more substantial. With similar lift being generated at both 90° and 75°, but with significantly higher drag at 90° compared to 75°, it is advantageous to install the propeller at a 75° angle to the wing rather than at 90°, especially at lower advance ratios. When increasing the size of the propeller, the effect on the WEP system also increases accordingly.

The linear superposition method, which involves adding the standalone propeller and wing performances, is feasible for estimating WEP system lift at large propeller incidence angles (greater than 60 degrees) with an average error of less than 12%, regardless of the propeller size with respect to chord. At higher advance ratios, the error increases to 20%. Also, the method underpredicts the lift generated by the WEP system when the propeller is placed at a 60° inclination angle due to the added positive propeller interference effect.

The linear superposition method is not suitable for estimating the forward thrust or drag experienced by the WEP system due to the additional drag generated by propeller-wing interaction. The error in forward thrust/drag estimation will also increase with advance ratio. The cause of the additional drag requires further aerodynamic investigations.

V.Acknowledgment

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VI.References

[1] Corgiat, A., Lind, G., Hartsel, J., “Fan-in-Wing Technology, from the XV-5A to the Present”. AIAA 1993.

[2] Deckert, W., “The Lift-Fan Aircraft: Lessons Learned”, 1995. NASA Ames Research Center, California.

[3] Gerdes, R., “Lift-Fan Aircraft: Lessons Learned from XV-5 Flight Experience”. AIAA 1993.

[4] AgustaWestland Project Zero. Retrieved from: leonardocompany.com

[5] Heyson, H. H., “Theoretical and experimental investigation of the performance of a fan-in-wing VTOL configuration”, NASA TN-D 7498, December 1973.

[6] Gregory, N., Raymer, W. G., and Love, E. M., “The Effect of Forward Speed on the Inlet Flow Distribution and Performance of a Lifting Fan Installed in a Wing,” Ministry of Aviation, Aeronautical Research Council Reports and Memoranda No. 3388, June 1962.

[7] Schaub, U. W., “Experimental Investigation of Flow Distortion in Fan-in-Wing Inlets,” Journal of Aircraft, Vol.5, No.5, pp 473-478, September 1968.

[8] Lieblein, S., Yuska, J. A., and Diedrich, J. H., “Performance Characteristics of a Model VTOL Lift Fan in Crossflow,” Journal of Aircraft, Vol.10, No.3, pp 131-136, March, 1973.

[9] Thouault, N., Breitsamter, C., and Adams, N. A., “Numerical and Experimental Analysis of a Generic Fan-in-Wing Configuration,” Journal of Aircraft, Vol 46. No. 2, pp 656-666, March 2009.

[10] Hickey, D., and Ellis, D., “Wind-Tunnel Tests of a Semispan Wing with a Fan Rotating in the Plane of the Wing”, 1959. NASA Ames Research Center, Washington.

[11] Culpepper, G, G., Cai, J., and Gunasekaran, S., “Propeller and Propeller-in-Wing Thrust Vectoring” SciTech 2021, AIAA 2021-0615.

[12] Kirk, J. V., Hall, L. P., and Hodder, B. K., “Aerodynamics of Lift Fan V/STPL Aircraft,” AIAA 8th Annual Metting and Technical Display, Washington, D.C, October 1971, AIAA 71-98.

[13] Culpepper, G, G., Cai, J., and Gunasekaran, S., “Effect of Forward Propeller Tilt and Inlet Shape in Propeller-in-Wing Thrust Vectoring” AIAA SciTech 2022, AIAA 2022-1539, doi: 10.2514/6.2022-1539

[14] Gogidze, G., Cai, J., and Gunasekaran, S., “Effect of Propeller Incidence Angle on Wing Embedded Propeller Configuration in Forward Flight” AIAA SciTech 2023, National Harbor, MD, Jan 2023, AIAA 2023-2284.

[15] ATI, F/T Gamma Sensor, Retrieved from: www.ati-ia.com.

[16] MathWorks, Designfilt, FIR band stop filter: https://www.mathworks.com/. [retrieved Feb. 2019]

[17] DAR Corporation, Flightstream, Retrieved from: www.darcorp.com/flightstream-aerodynamics-software

[18] McCormick, B. W., "Aerodynamics, Aeronautics, and Flight Mechanics (Second Edition)," John Wiley & Sons, New York, 1995, p. 116.